WEBERMAN RESIDENCE

Design Development Concept Construction Set

Mountain courtyard house, Thailand · Owner: Max Weberman · Issue DD-01, 11 Oct 2026

DESIGN DEVELOPMENT, NOT FOR CONSTRUCTION

SHEET INDEX

SHEET	TITLE	SCALE
G-001	Cover Sheet + Sheet Index	As Noted
G-002	Sustainability Summary	NTS
G-003	Construction Standards (Thai vs Western)	NTS
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A-102	Roof Plan	1/8" = 1'-0"
A-201	Exterior Elevations	1/8" = 1'-0"
A-301	Building Sections	3/16" = 1'-0"
A-401	Kitchen Plan + Elevations	1/4" = 1'-0"
S-101	Foundation + Structural Plan	1/8" = 1'-0"
S-102	Wall Framing + Roof Framing Plans	3/32" = 1'-0"
E-101	Power + Lighting Plan	1/8" = 1'-0"
P-101	Plumbing + Site Utilities Plan	3/32" = 1'-0"
M-101	HVAC Plan	1/8" = 1'-0"
E-102	Solar PV Plan + Single-Line Diagram	3/32" = 1'-0"
E-103	Controls + Monitoring	NTS
P-102	Rainwater, Irrigation + Kitchen Garden Plan	1/16" = 1'-0"
P-103	Water Source + Treatment	NTS

ONE SHARED MODEL

Every sheet in this set and the interactive 3D model are generated from one data model: plan.json (rooms, walls, openings, roofs, site) extended into bim.json with 1,512 elements in twelve layers (Architecture 82, Structure 369, Framing 142, Roof 153, Electrical 383, Plumbing 160, HVAC 68, Site Utilities 10, Water + Rainwater 29, Solar 48, Controls 46, Landscape + Farm 22). Change the model and both regenerate, so drawings and 3D always match.

PROJECT DATA

Conditioned area	5,272 SF (490 m²), single level
Footprint	94' x 83' around a 36' x 28' courtyard
Structure	RC frame on spread footings, AAC + LGS walls, steel roof framing
Roofs	Standing seam: butterfly (Living) + 5 shed roofs, 4-8 ft eaves
Services	PEA 3-phase + solar/battery/generator; well + rainwater; septic + ATU; saltwater pool
Codes (to confirm)	Building Control Act B.E. 2522, Ministerial Regs, EIT electrical standard, DPT 1301/1302/1311



K **KEY PLAN**

SCALE: 1/32" = 1'-0"

ASSUMPTIONS FOR THAI ENGINEER VERIFICATION

1. Allowable soil bearing 10 t/m² and footing depth 1.5 m (no soil test yet). Mountain slope may require piles or stepped footings.
2. RC member sizes (C1 250x250, GB1 200x500, RB1 200x400, S1 125 mm) and rebar are placeholders for a 1-storey frame at about 4.5 m bays.
3. Steel transfer beams TB1 over the 21 ft pocketing glass and frameless corner: assumed W250, deflection limit L/600 for glass.
4. Steel rafters H-150 cantilevering 4-8 ft at eaves; uplift and connection design by engineer (DPT 1311 wind).
5. Electrical: 3-phase 400/230 V PEA service, 125 A main; load calc, cable sizes and RCBO selection to EIT standard at CD.
6. Cooling at about 850 BTU/h per m²; VRF/multi-split outdoor units CU-1/2/3 sized by connected load, not by a heat-load calc.
7. Plumbing pipe sizes by fixture count rule of thumb; septic + ATU and drain field sized for 6 people, perc test required.
8. Rainwater ring, tanks (2 x 19,000 L), well and pool routes are schematic; final routes per survey and civil grading.
9. Saltwater pool: corrosion-resistant fittings, backwash to lined soakaway; bonding/equipotential grid per EIT.

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SCALE
As Noted

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

GENERAL NOTES

Concept design development set for coordination and pricing. Not for permit or construction.

Model coordinates: feet, x east, y north, FFL = 0.

Interactive 3D: webermanapps.github.io/weberman-residence (Layers panel).

SHEET TITLE
Cover Sheet + Sheet Index

SHEET
G-001

SUSTAINABLE LUXURY

Quiet, high-craft living that makes its own energy and water and closes the food + waste loop.

20.0 kWp
Solar PV + 30 kWh battery

~77%
Electricity from the sun

80 m³
Rainwater storage for the garden

~75%
Dry-season garden water from rain

Well
Primary house water; mains backup only

4-6 Hens
Eggs + compost loop

MOVES, STATUS + ROUGH COST (THB)

MOVE	STATUS	ROUGH THB	IMPACT
Rooftop solar 20 kWp + 30 kWh battery (E-102)	Included	1,020,000	~77% of electricity, ~13 t CO2/yr avoided
Rainwater cisterns 80 m³ + smart drip (P-102)	Included	710,000	~75% of dry-season garden water
Drilled well + treatment, mains as backup only (P-103)	Included	561,000	Water self-sufficiency
Home controller: solar-timed pool/irrigation, leak shut-off (E-103)	Included	347,000	10-15% energy, leak protection
Heat-pump water heater 300 L	Included	85,000	~3x less energy than resistive
Passive cooling: deep eaves, courtyard cross-ventilation, ceiling fans, light roofs	Included	90,000	Cuts AC hours 20-30%
ERV fresh air with enthalpy core	Included	180,000	Healthy air at low cooling cost
Greywater reuse to subsurface drip (GW-1)	Included	60,000	~150-250 L/day reused
Kitchen garden, raised beds, compost + 4-6 hens	Included	160,000	Food + closes the waste loop
Indoor hydroponic herb nook (solar-powered LEDs)	Included	589,000	Year-round herbs, zero food miles
EV-ready: 7.4 kW charger + spare conduit to carport	Included	45,000	Solar-charged driving
Low-carbon materials: AAC block, slag/fly-ash cement blend, plantation teak, charred timber (no paint), lime plaster	Included	Cost-neutral	~15-25% less embodied carbon (cost-neutral)
Bidirectional EV charger (V2H backup)	Optional	350,000	Car becomes a 60 kWh home battery
Solar-thermal / heat-pump pool heating on surplus	Optional	120,000	Warm pool without grid power
Green (sedum) roof on studio, + acoustic + cooling	Optional	250,000	Cooler studio, habitat
Biogas digester for kitchen + coop waste	Optional	40,000	Cooking gas, fertilizer
Earth-tube pre-cooling for ERV intake	Optional	150,000	2-4 °C cooler fresh air

Included total about 3,847,000 THB · Optional add-ons about 910,000 THB (concept-level, 2026 prices, excl. VAT)

Sheet references: E-102 Solar, E-103 Controls, P-102 Rainwater + Kitchen Garden, P-103 Well + Treatment, M-101 HVAC. In the 3D viewer, the Solar, Water + Rainwater, Controls and Landscape + Farm layers show each system in place.

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NOTES
Costs are rough order-of-magnitude for budgeting; confirm with Thai contractors.
Carbon figures use the Thai grid factor ~0.5 kg CO2/kWh.

SHEET TITLE
Sustainability Summary

SHEET
G-002

CONSTRUCTION STANDARDS: THAI VS WESTERN

Approved project standard (Oct 11 2026). Applies to every sheet and to the element properties in the 3D model.

TOPIC	COMMON THAI PRACTICE	OUR STANDARD	WHY
Structure	RC frame with brick infill; timber roof/floor framing common	RC frame (columns, beams, slabs); steel roof framing; no timber framing	Termites and humidity destroy timber; RC + steel is durable and insurable
Exterior Walls	Single-skin red clay brick or thin block, cement render, no insulation	200 mm AAC + exterior insulation or ventilated charred-timber cladding; lime plaster	Cuts heat gain and AC load; breathable lime resists mould
Interior Partitions	Clay brick / block, rendered	Light-gauge steel studs with board; acoustic wool at studio + bedrooms	Lighter, straighter, quieter; termite-proof; easy service routing
Concrete Walls	Poured where convenient	Only pool, retaining, cistern, engineer-required shear walls	Less embodied carbon and heat mass where not needed
Roof	Metal sheet or tile on timber/steel, foil at best, little insulation	Metal over radiant barrier, 75 mm insulation, ventilated cavity	Hot-roof heat is the #1 cooling load; quiet in monsoon rain
Traps	Bottle traps or none; floor drains often untrapped (smell)	P-trap on every fixture; deep-seal trapped floor drains with trap primers	Stops sewer gas, insects and odours
Venting	Often unvented; drains glug and siphon traps	Proper vent stack system through the roof	Keeps trap seals; fast, quiet drainage
Grease	Kitchen waste straight to sewer/septic	Kitchen grease interceptor GT-1 before septic	Protects septic + ATU and drain field
Water Supply	PVC cold + mixed fittings, rarely tested	PPR fusion-welded, pressure-tested before walls close	No hidden leaks; rated for hot water
Electrical	Partial earthing, RCD on some circuits, ad-hoc panel	TN-C-S earthing, 30 mA RCBO on all circuits, full panel schedule	Shock + fire safety in a wet climate
Wet Areas	Tile on render; leaks common	Waterproofing membrane in all wet areas, flood-tested 24 h	Prevents leaks and mould into AAC + studs
Hot Water	Point-of-use electric heaters at each shower	Central heat-pump water heater + recirculation	~3x more efficient; one serviceable unit; instant hot water

WALL TYPES

- W1 AAC Exterior: 200 mm AAC block in RC frame; exterior insulation or ventilated charred-timber cladding; lime plaster
- W2 Steel Stud: 92 mm LGS @ 400 mm, board both sides, mineral wool; cement board + membrane on wet faces
- W3 Acoustic Stud: Double/isolated LGS, 75 mm acoustic wool, 2 x sound board, resilient channel (studio + bedrooms)
- W4 RC Wall: Poured RC: pool, retaining walls, cistern, engineer-required shear walls only

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NOTES

Western standard = trapped + vented DWV, tested PPR, full earthing + RCBO, membranes, central hot water.

Engineer to confirm shear walls (W4) and all member sizes.

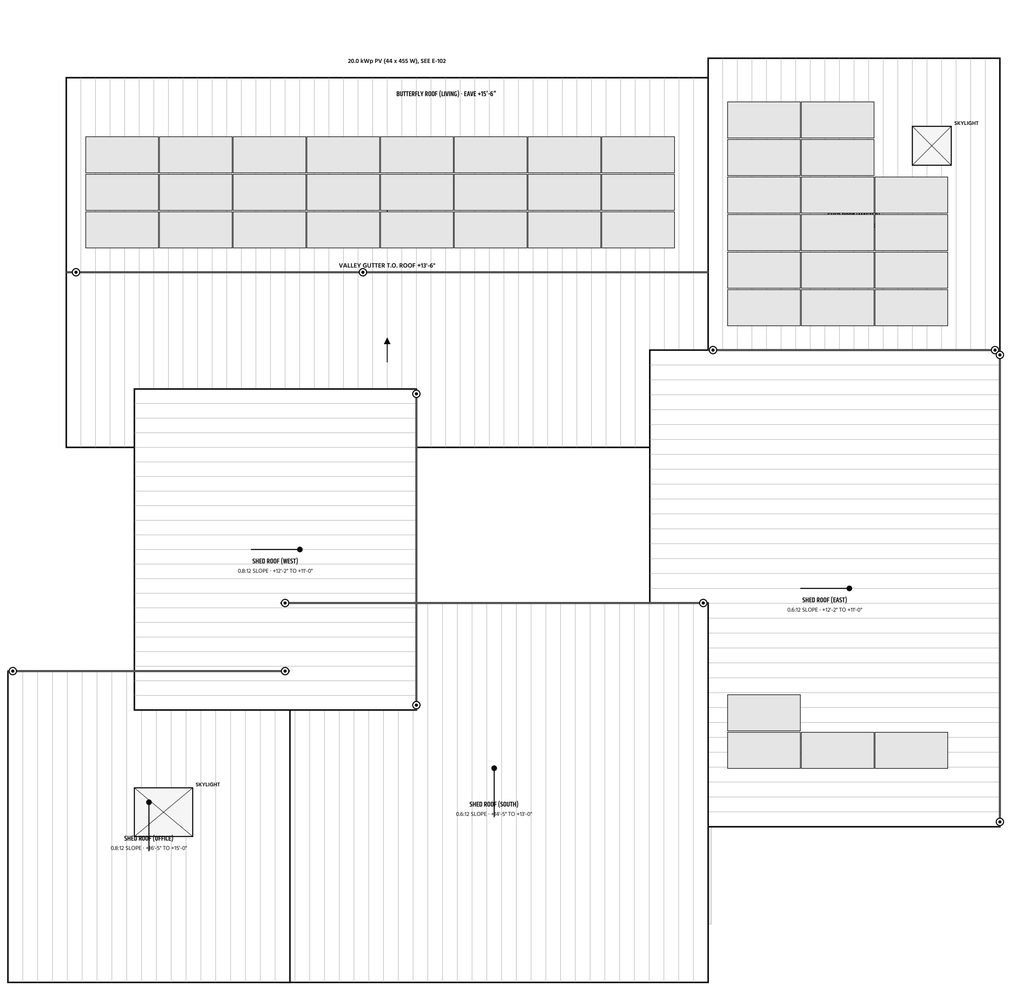
SHEET TITLE

Construction Standards (Thai vs Western)

SHEET

G-003


94'-0" (28.65 m) OVERALL
10'-0"
16'-0"
14'-0"
22'-0"
14'-0"
12'-0"
HERB NOOK
W102
W104
D02
D02
W05
W06
W05
W07
W08
W09
W10
W11
W12
W13
W14
W15
W16
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W37
W38
W39
W40
W41
W42
W43
W44
W45
W46



ROOF LEGEND

- Gutter, 150 x 125 mm AZ steel
- Downspout / rain chain to buried leader
- Valley gutter
- Standing seam at 450 mm
- Slope (dot = low side)

1 **ROOF PLAN**

SCALE: 1/8" = 1'-0"

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SCALE
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ROOF NOTES

Standing seam 0.5 mm AZ150, 75 mm PIR, vapor-open underlay on steel C-purlins (S-102).

Deep eaves 4-8 ft: cantilevered rafters; confirm for 0.6-1.0 kPa wind uplift (T15 / DPT 1.11).

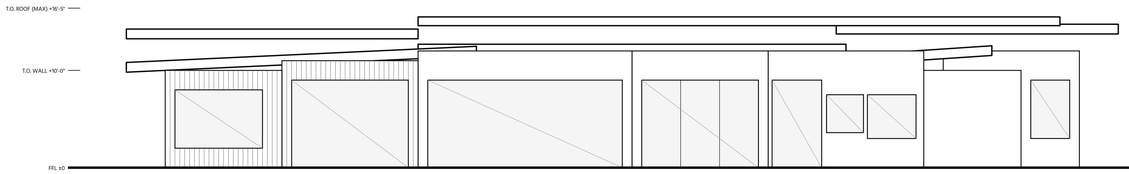
All roofs drain to the rainwater ring (P-101) and two 19,000 L tanks.

SHEET TITLE

Roof Plan

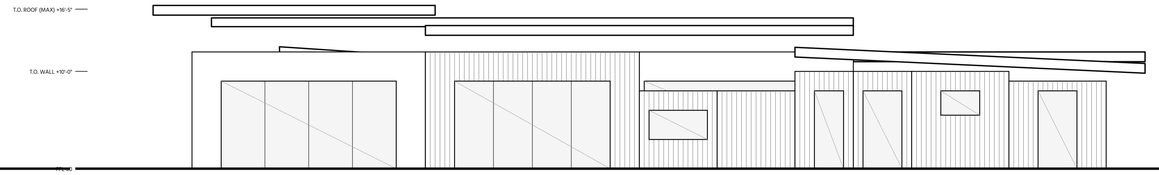
SHEET

A-102



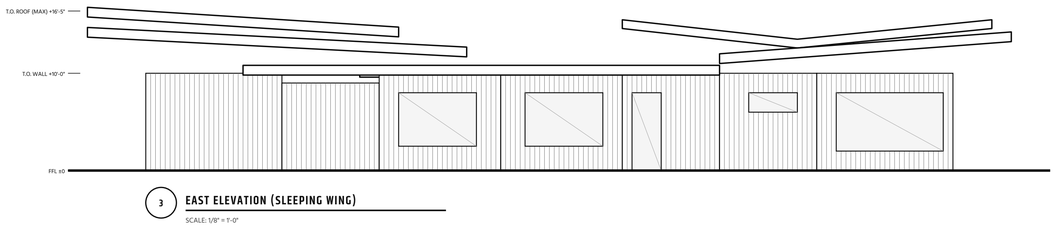
1 **NORTH ELEVATION (MOUNTAIN SIDE)**

SCALE: 1/8" = 1'-0"



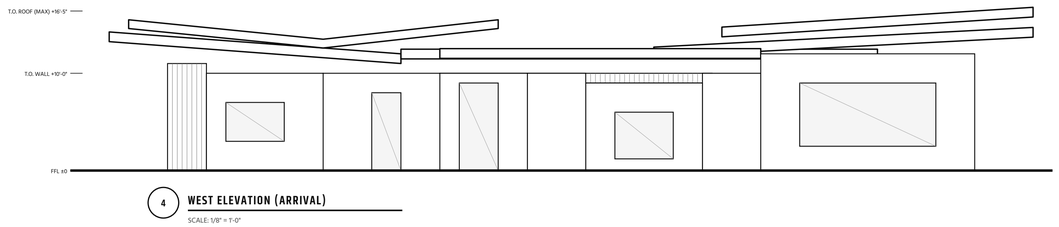
2 **SOUTH ELEVATION (POOL SIDE)**

SCALE: 1/8" = 1'-0"



3 **EAST ELEVATION (SLEEPING WING)**

SCALE: 1/8" = 1'-0"



4 **WEST ELEVATION (ARRIVAL)**

SCALE: 1/8" = 1'-0"

MATERIAL LEGEND

- White lime plaster with black timber trim
- Charred timber (yakisugi) siding
- Bronze-framed glazing / doors
- Standing-seam roof, charred fascia

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ELEVATION NOTES

Elevations generated from the shared model; walls are drawn to 12 ft max where they continue up under the roof.

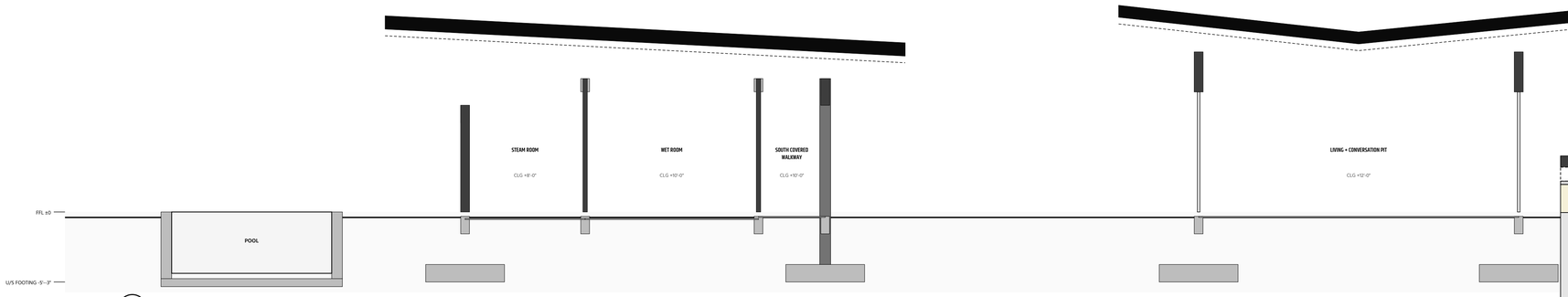
Grade shown level; final site grading and retaining by civil engineer.

SHEET TITLE

Exterior Elevations

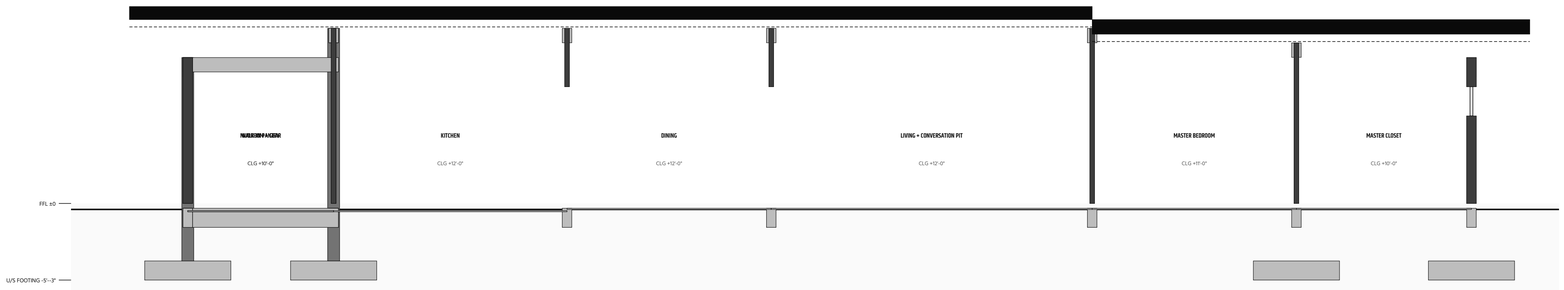
SHEET

A-201



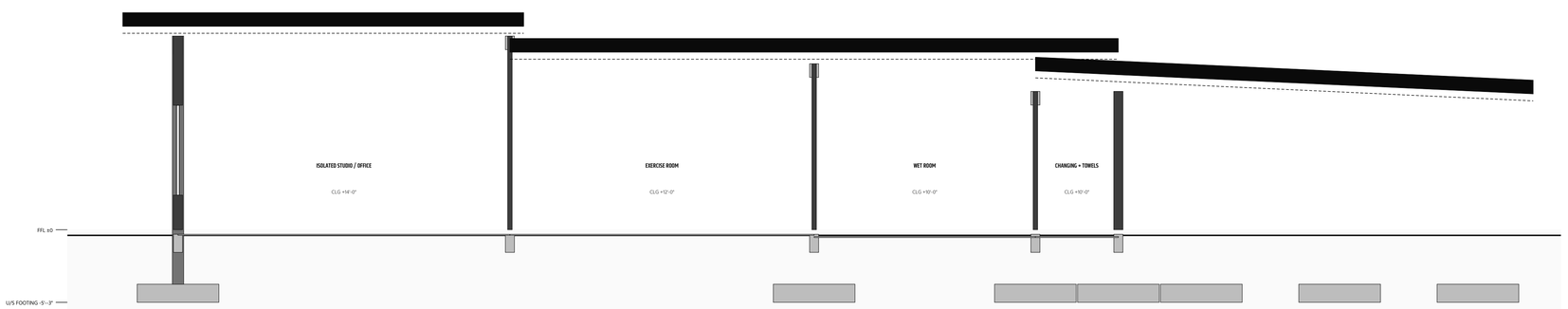
1 SECTION A-A (NORTH-SOUTH THROUGH LIVING, COURTYARD, WET ROOM, POOL)

SCALE: 3/16" = 1'-0"



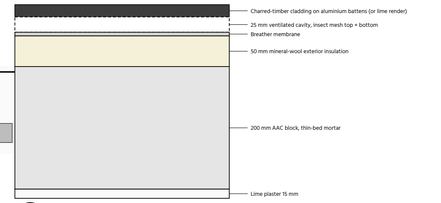
2 SECTION B-B (EAST-WEST THROUGH KITCHEN, DINING, LIVING, MASTER)

SCALE: 3/16" = 1'-0"



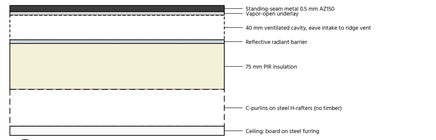
3 SECTION C-C (EAST-WEST THROUGH STUDIO, GYM, WET ROOM, CHANGING)

SCALE: 3/16" = 1'-0"



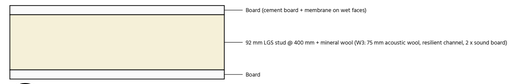
4 TYPICAL EXTERIOR WALL W1 (PLAN CUT, OUT TO IN)

SCALE: 1:10 (NTS)



5 TYPICAL ROOF (TOP DOWN)

SCALE: 1:10 (NTS)



6 TYPICAL PARTITION W2 / W3

SCALE: 1:10 (NTS)

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REVISION
DD-01

SECTION NOTES

Section cuts at x = 60 ft (A), y = 67 ft (B) and y = 11 ft (C) in model coordinates; see key on S-101.

Footing depth 1.5–1.6 m assumed pending soil investigation.

Black = roof assembly; dashed = steel rafter zone; dark gray = walls cut; light gray = RC.

STANDARD

RC frame + AAC + LGS + steel roof; no timber framing.

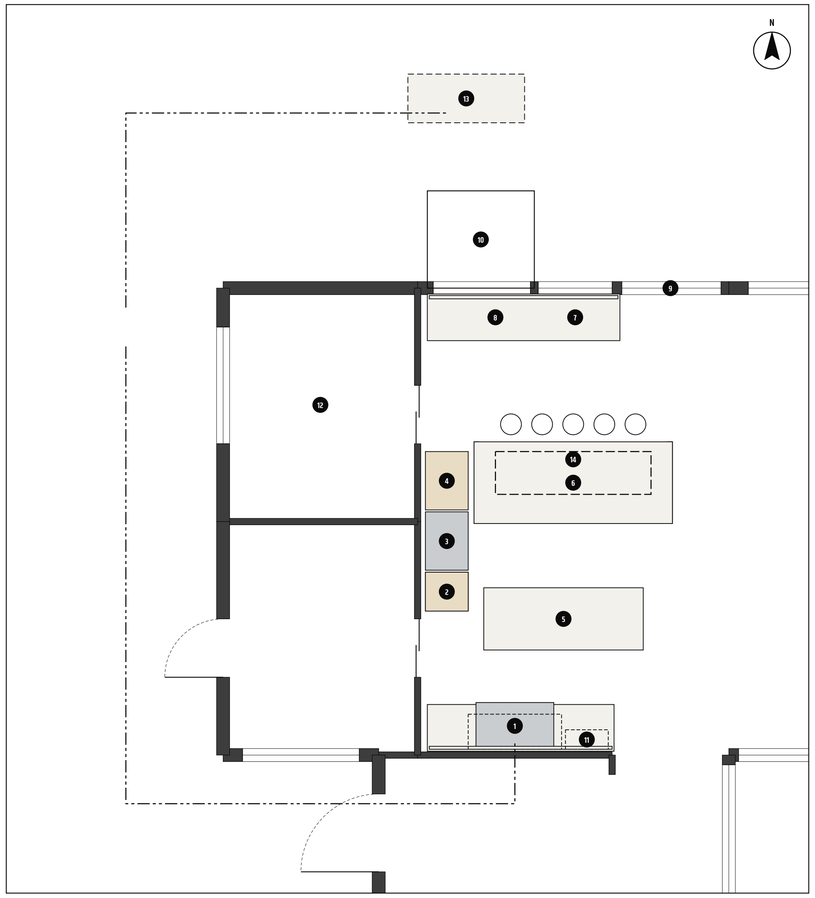
Wet areas: membrane, flood test 24 h.

SHEET TITLE

Building Sections

SHEET

A-301

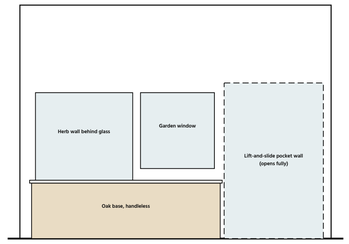


1 KITCHEN PLAN

SCALE: 1/4" = 1'-0"

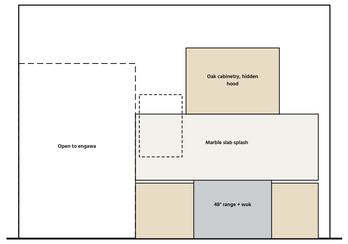
KEYNOTES

1. 48" pro gas range, wok burner, hood hidden in oak cabinetry above
2. Appliance garage, oak pocket doors, GFCI outlets
3. Built-in stainless fridge 36" (not panel-ready)
4. Steam oven + wall oven column
5. Work island: flush 2-zone induction, quartzite top
6. 10' quartzite waterfall island, 5 stools on one side
7. Integrated quartzite sink + brushed stainless faucet, garden-view window
8. Dishwasher left of sink
9. Lift-and-slide pocket wall, opens fully to the pergola
10. Hydroponic herb wall (glazed nook) behind glass backsplash
11. Open rift oak shelves
12. Walk-in pantry (104), no scullery
13. Optional covered outdoor wok station
14. Skylight SK-1 over island + linear suspended LED



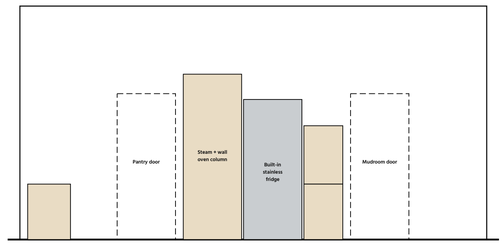
2 NORTH ELEVATION: SINK + SLIDE WALL

SCALE: 1/4" = 1'-0"



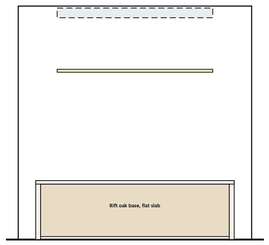
3 SOUTH ELEVATION: RANGE WALL

SCALE: 1/4" = 1'-0"



4 WEST ELEVATION: TALL WALL + PANTRY

SCALE: 1/4" = 1'-0"



5 ISLAND ELEVATION: QUARTZITE WATERFALL

SCALE: 1/4" = 1'-0"

MATERIAL + APPLIANCE SCHEDULE

ITEM	SELECTION
Cabinets	Rift white oak, flat slab, handleless (finger-pull reveal)
Counters	Honed quartzite, 30 mm, mitred 60 mm edge
Island	10' quartzite waterfall, stools on one side
Sink	Integrated quartzite, carved from the counter
Faucet	Brushed stainless (not brass, kitchen only)
Backsplash	Honed marble slab (zellige or plaster alternate)
Range	48" pro gas with high-heat wok burner
Hood	Insert hidden in oak cabinetry, 1,500 m3/h + MUA
Fridge	36" built-in stainless
Ovens	Steam oven + wall oven column
Cooktop 2	Flush 2-zone induction in work island
Lighting	Skylight + linear suspended LED; shelf strips
Opening	Lift-and-slide pocket wall to pergola

ROUGH COST (THB, EXCL. VAT)

ITEM	THB
Rift white oak flat-slab handleless cabinetry (~11 m run + 2 islands + tall wall)	1,350,000
Quartzite slabs: counters, waterfall island, integrated sink	950,000
48" pro gas range with wok burner	420,000
Hidden hood insert + 1,500 m³/h blower + MUA-1	220,000
Steam oven + wall oven column	330,000
Built-in stainless fridge 36"	360,000
Flush 2-zone induction (work island)	85,000
Skylight SK-1 + linear suspended LED	210,000
Lift-and-slide pocket wall (5 m, bronze)	520,000
LPG cage, gas line, solenoid + detector	65,000
Outdoor wok station (optional)	120,000

Kitchen total about 4,650,000 THB

Excluded per swipe: scullery/back kitchen, table-extension or furniture-style island, limestone, terrazzo, walnut, painted or plaster cabinets, shaker doors, undermount stainless sink, unlacquered brass faucet, panel-ready fridge.

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KITCHEN NOTES

Wok burner stays in the main kitchen (no scullery) with a wok-rated hidden hood and interlocked make-up air; outdoor wok station is optional.

Gas leak detector closes the range solenoid and boosts the hood (E-103).

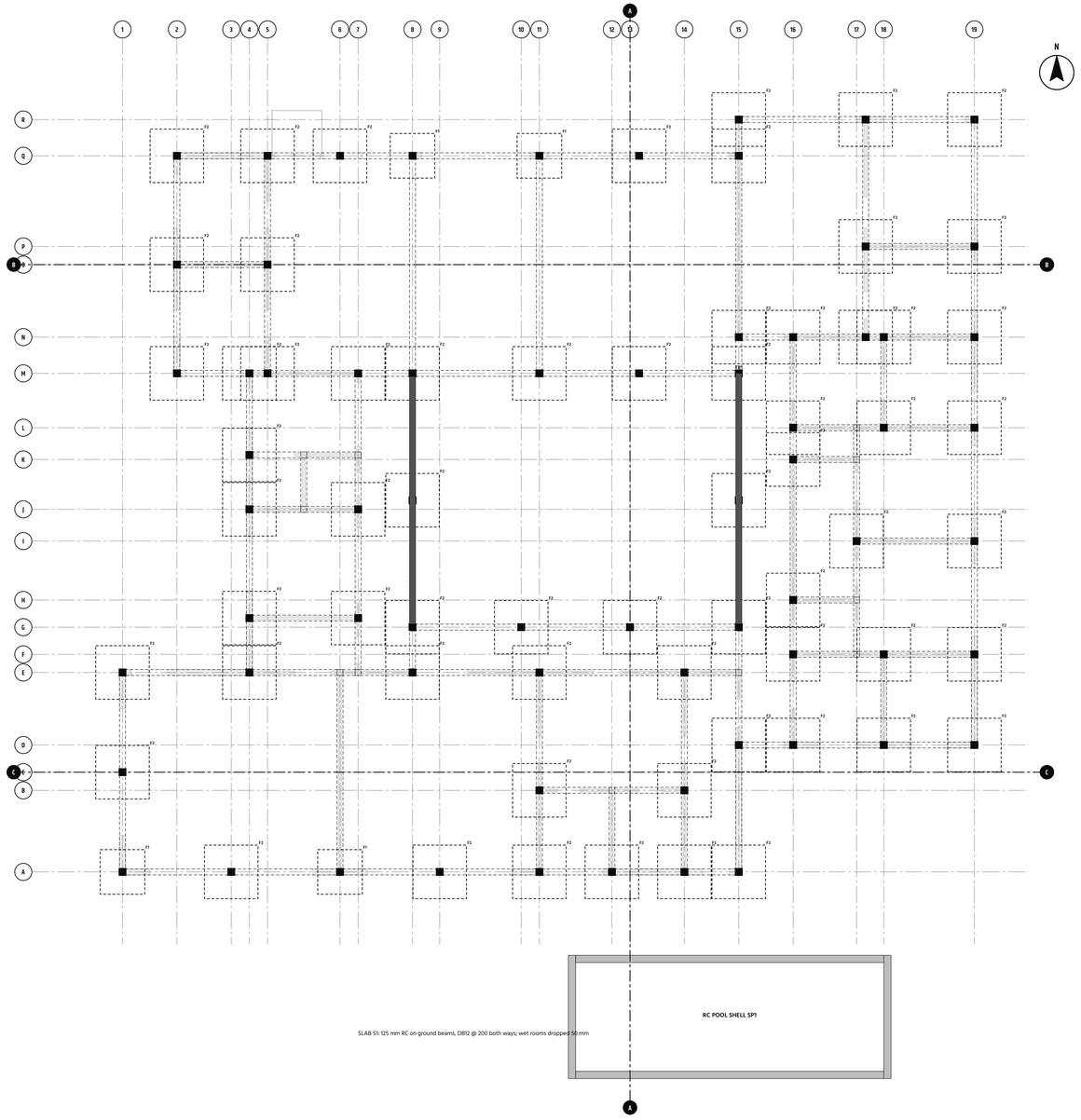
Clearances: 42" min between islands and runs.

SHEET TITLE

Kitchen Plan + Elevations

SHEET

A-401



STRUCTURAL SCHEDULE

MARK	ELEMENT	SIZE (MM)	REINFORCEMENT (INDICATIVE)	QTY
F1	Spread footing (corner / edge)	1,500 x 1,500 x 400	10-DB16 each way, 75 mm cover	4
F2	Spread footing (interior / multi-bay)	1,800 x 1,800 x 400	12-DB16 each way	64
C1	RC column	250 x 250	4-DB16 (8-DB16 at TB1), RB9 ties @ 150	68
GB1	Ground beam	200 x 500	3-DB16 top + bot, RB9 @ 150	100
RB1	Roof ring beam (exterior)	200 x 400	2-DB16 top + bot, RB9 @ 150	37
RB2	Roof beam (interior bearing)	150 x 300	2-DB12 top + bot, RB6 @ 150	64
TB1	Steel transfer beam over wide glazing	W250 assumed	Engineer to size, deflection L/600 for glass	2
S1	Suspended floor slab	125 thick	DB12 @ 200 B.W., WP membrane in wet rooms	31
SP1	Pool shell	250 walls + floor	Waterproof concrete, DB16 @ 150 E.F.	1

STRUCTURAL LEGEND

- RC column C1
- Footing (below)
- Ground beam GB1 (below)
- Structural grid

1 **FOUNDATION + STRUCTURAL PLAN**

SCALE: 1/8" = 1'-0"

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
1/8" = 1'-0"

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

STRUCTURAL NOTES (INDICATIVE)

Concrete: f'c' 240 ksc (cylinder) footings + frame; 280 ksc pool. Rebar: SD40 deformed, SR24 round.

Allowable soil bearing assumed 10 t/m² pending a soil test; footings 1.5 m below grade.

Seismic zone + wind per DPT 1301/1302 and 1311 to be confirmed by a licensed Thai civil engineer (Saman Witsawakon).

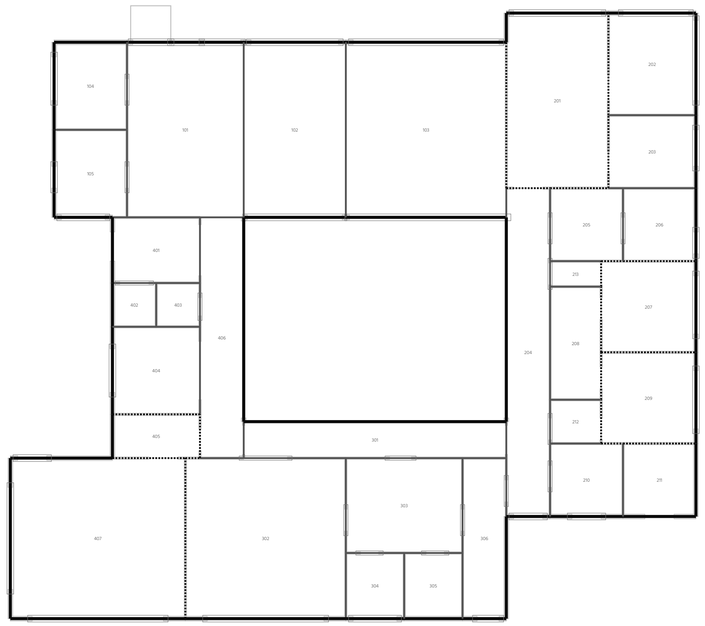
All sizes are concept placeholders for coordination only.

SHEET TITLE

Foundation + Structural Plan

SHEET

S-101



1 WALL FRAMING PLAN

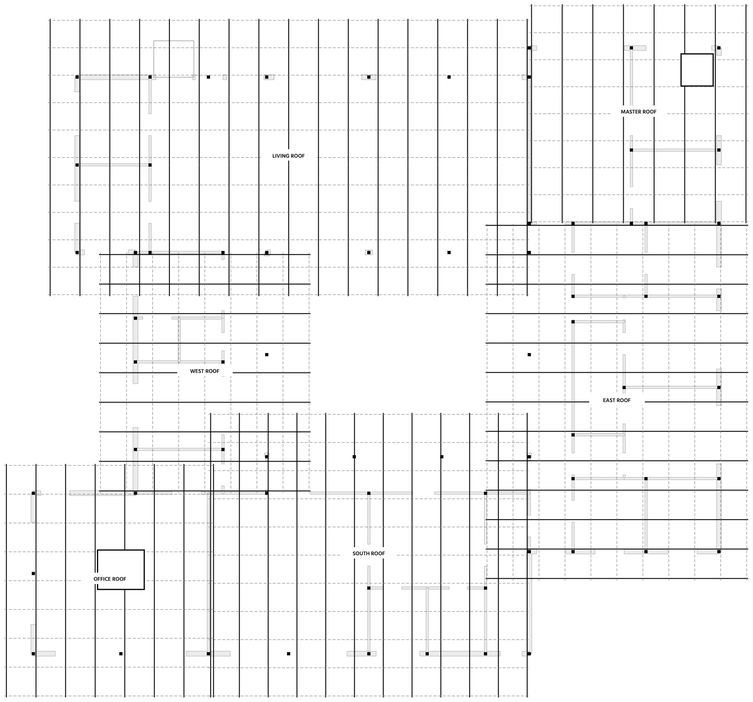
SCALE: 3/32" = 1'-0"

WALL TYPES

TYPE	DESCRIPTION	BUILD-UP	LENGTH
W1	AAC Block Exterior Wall	200 mm AAC block in RC frame (RC lintels + stiffeners), 50 mm mineral-wool exterior insulation OR ventilated charred-timber cladding on aluminium battens, lime plaster both faces	498 ft
W2	Steel Stud Partition (Wet Face)	92 mm LGS @ 400 mm, cement board on wet face + liquid/sheet waterproofing membrane (min 1.8 m in showers, full height in steam), gypsum on dry face	407 ft
W3	Acoustic Steel Stud Partition	92 mm LGS @ 400 mm (double stud on isolated tracks at studio 407), 75 mm acoustic mineral wool, 2 x 13 mm sound-rated board each side, resilient channel (STC 55+, 60+ at studio)	174 ft

WALL LEGEND

- W1 AAC exterior wall 200 mm
- W2 LGS steel stud partition
- W3 Acoustic steel stud (studio + bedrooms)
- W4 RC wall (pool, retaining, cistern, shear)
- L1 RC lintel over opening



2 ROOF FRAMING PLAN

SCALE: 3/32" = 1'-0"

ROOF FRAMING SCHEDULE

MARK	MEMBER	SECTION / SPEC	QTY
R1	Steel rafter	H-150 x 75 x 5 x 7, SS400, @ 1.2 m, cantilevers to eaves	84
P1	C-purlin	C-100 x 50 x 20 x 2.3, @ 1.1 m max	60
SK	Skylight trimmer	RHS 100 x 50 x 3.2 frame	4
--	Roof deck	Standing seam 0.5 mm AZ150 + 75 mm PIR + underlay	6
--	Connections	Rafter to RB1 via cast-in plate 200 x 200 x 12 + 4-M16 anchors	-
--	Corrosion	Hot-dip galvanized or 2-coat epoxy (humid mountain climate)	-

ROOF FRAMING LEGEND

- Steel rafter R1
- C-purlin P1
- Column below (bearing)

WALL TYPES (APPROVED STANDARD)

- W1 AAC Exterior: 200 mm AAC block in RC frame; exterior insulation or ventilated charred-timber cladding; lime plaster
- W2 Steel Stud: 92 mm LGS @ 400 mm, board both sides, mineral wool; cement board + membrane on wet faces
- W3 Acoustic Stud: Double/isolated LGS, 75 mm acoustic wool, 2 x sound board, resilient channel (studio + bedrooms)
- W4 RC Wall: Poured RC: pool, retaining walls, cistern, engineer-required shear walls only

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

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MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

FRAMING NOTES

Interior LGS: 0.75 mm BMT studs, deflection head track at slab/beam.

AAC: thin-bed mortar; RC stiffener columns 100 x wall width at max 3 m and at openings > 2 m.

Studio double-stud walls on separate tracks + resilient ceiling (STC 60+).

STANDARD

No timber framing (termites): RC frame, AAC, LGS studs, steel roof.

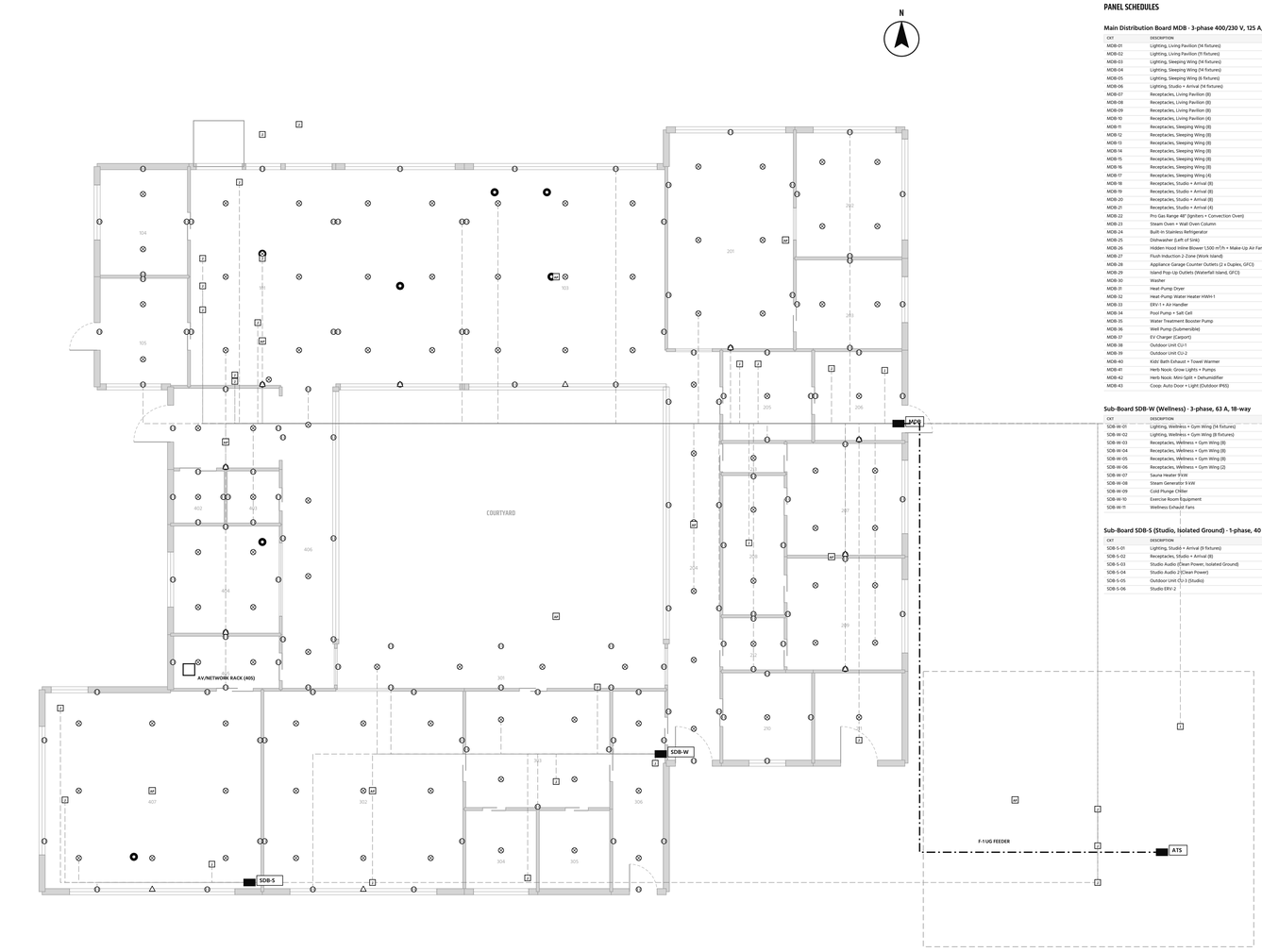
Poured concrete only at pool, retaining walls, cistern and engineer-required shear walls (W4).

SHEET TITLE

Wall Framing + Roof Framing Plans

SHEET

S-102



ELECTRICAL LEGEND

- Duplex receptacle 16 A (RCBO in wet areas)
- LED downlight 2700 K dimmable
- Decorative pendant / lantern
- Dedicated circuit / equipment connection
- Data outlet 2 x Cat6A
- Wi-Fi access point (PoE)
- Distribution board
- Circuit home run (conduit in ceiling)
- Underground feeder

PANEL SCHEDULES

Main Distribution Board MDB · 3-phase 400/230 V, 125 A, 36-way, RCBO protected

CKT	DESCRIPTION	LOAD	BREAKER
MDB-01	Lighting, Living Pavilion (14 fixtures)	140 W	10 A RCBO
MDB-02	Lighting, Living Pavilion (11 fixtures)	110 W	10 A RCBO
MDB-03	Lighting, Sleeping Wing (14 fixtures)	140 W	10 A RCBO
MDB-04	Lighting, Sleeping Wing (14 fixtures)	140 W	10 A RCBO
MDB-05	Lighting, Sleeping Wing (6 fixtures)	60 W	10 A RCBO
MDB-06	Lighting, Studio + Arrival (14 fixtures)	140 W	10 A RCBO
MDB-07	Receptacles, Living Pavilion (8)	1,800 VA	16 A RCBO
MDB-08	Receptacles, Living Pavilion (8)	1,800 VA	16 A RCBO
MDB-09	Receptacles, Living Pavilion (8)	1,800 VA	16 A RCBO
MDB-10	Receptacles, Living Pavilion (4)	1,800 VA	16 A RCBO
MDB-11	Receptacles, Sleeping Wing (8)	1,800 VA	16 A RCBO
MDB-12	Receptacles, Sleeping Wing (8)	1,800 VA	16 A RCBO
MDB-13	Receptacles, Sleeping Wing (8)	1,800 VA	16 A RCBO
MDB-14	Receptacles, Sleeping Wing (8)	1,800 VA	16 A RCBO
MDB-15	Receptacles, Sleeping Wing (8)	1,800 VA	16 A RCBO
MDB-16	Receptacles, Sleeping Wing (8)	1,800 VA	16 A RCBO
MDB-17	Receptacles, Sleeping Wing (4)	1,800 VA	16 A RCBO
MDB-18	Receptacles, Studio + Arrival (8)	1,800 VA	16 A RCBO
MDB-19	Receptacles, Studio + Arrival (8)	1,800 VA	16 A RCBO
MDB-20	Receptacles, Studio + Arrival (8)	1,800 VA	16 A RCBO
MDB-21	Receptacles, Studio + Arrival (4)	1,800 VA	16 A RCBO
MDB-22	Pro Gas Range 48" (Ignites + Convection Oven)	18 kW	16 A
MDB-23	Steam Oven + Wall Oven Column	6.5 kW	32 A 2P
MDB-24	Built-In Stainless Refrigerator	0.6 kW	16 A
MDB-25	Dishwasher (Left of Sink)	1.8 kW	16 A
MDB-26	Hidden Hood Inline Blower 1,500 m³/h + Make-Up Air Fan	0.8 kW	16 A
MDB-27	Flush Induction 2-Zone (Work Island)	3.7 kW	20 A
MDB-28	Appliance Garage Counter Outlets (2 x Duplex, GFCI)	1.8 kW	16 A RCBO
MDB-29	Island Pop-Up Outlets (Waterfall Island, GFCI)	1.8 kW	16 A RCBO
MDB-30	Washer	2.2 kW	16 A
MDB-31	Heat-Pump Dryer	1.2 kW	16 A
MDB-32	Heat-Pump Water Heater HWH-1	1.5 kW	16 A
MDB-33	ERV-1 + Air Handler	0.4 kW	10 A
MDB-34	Pool Pump + Salt Cell	1.5 kW	16 A
MDB-35	Water Treatment Booster Pump	1.1 kW	16 A
MDB-36	Well Pump (Submersible)	1.5 kW	16 A
MDB-37	EV Charger (Carport)	7.4 kW	32 A 2P
MDB-38	Outdoor Unit CU-1	9.0 kW	3P 25 A
MDB-39	Outdoor Unit CU-2	12.0 kW	3P 25 A
MDB-40	Kids' Bath Exhaust + Towel Warmer	0.2 kW	10 A
MDB-41	Herb Nook: Grow Lights + Pumps	1.2 kW	16 A RCBO
MDB-42	Herb Nook: Mini-Split + Dehumidifier	1.2 kW	16 A
MDB-43	Coop: Auto Door + Light (Outdoor IP65)	0.1 kW	10 A RCBO

Sub-Board SDB-W (Wellness) · 3-phase, 63 A, 18-way

CKT	DESCRIPTION	LOAD	BREAKER
SDB-W-01	Lighting, Wellness + Gym Wing (14 fixtures)	140 W	10 A RCBO
SDB-W-02	Lighting, Wellness + Gym Wing (8 fixtures)	80 W	10 A RCBO
SDB-W-03	Receptacles, Wellness + Gym Wing (8)	1,800 VA	16 A RCBO
SDB-W-04	Receptacles, Wellness + Gym Wing (8)	1,800 VA	16 A RCBO
SDB-W-05	Receptacles, Wellness + Gym Wing (8)	1,800 VA	16 A RCBO
SDB-W-06	Receptacles, Wellness + Gym Wing (2)	1,800 VA	16 A RCBO
SDB-W-07	Sauna Heater 9 kW	9.0 kW	3P 16 A
SDB-W-08	Steam Generator 9 kW	9.0 kW	3P 16 A
SDB-W-09	Cold Plunge Chiller	1.2 kW	16 A
SDB-W-10	Exercise Room Equipment	2.0 kW	16 A
SDB-W-11	Wellness Exhaust Fans	0.3 kW	10 A

Sub-Board SDB-S (Studio, Isolated Ground) · 1-phase, 40 A, 12-way

CKT	DESCRIPTION	LOAD	BREAKER
SDB-S-01	Lighting, Studio + Arrival (9 fixtures)	90 W	10 A RCBO
SDB-S-02	Receptacles, Studio + Arrival (8)	1,800 VA	16 A RCBO
SDB-S-03	Studio Audio (Clean Power, Isolated Ground)	2.0 kW	16 A IG
SDB-S-04	Studio Audio 2 (Clean Power)	2.0 kW	16 A IG
SDB-S-05	Outdoor Unit CU-3 (Studio)	4.5 kW	25 A
SDB-S-06	Studio ERV-2	0.2 kW	10 A

1 **POWER + LIGHTING PLAN**

SCALE: 1/8" = 1'-0"

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
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SCALE
1/8" = 1'-0"

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

ELECTRICAL NOTES

PEA 3-phase 400/230 V service; TN-C-S earthing per EIT standard, < 5 ohm.

RCBO (30 mA) on all receptacle, wet-area and outdoor circuits.

Light switch style is an open item (Round 2); switching and dimming zones to be shown at CD.

Wire sizes per EIT 2001-56 at CD; load calc pending equipment selection.

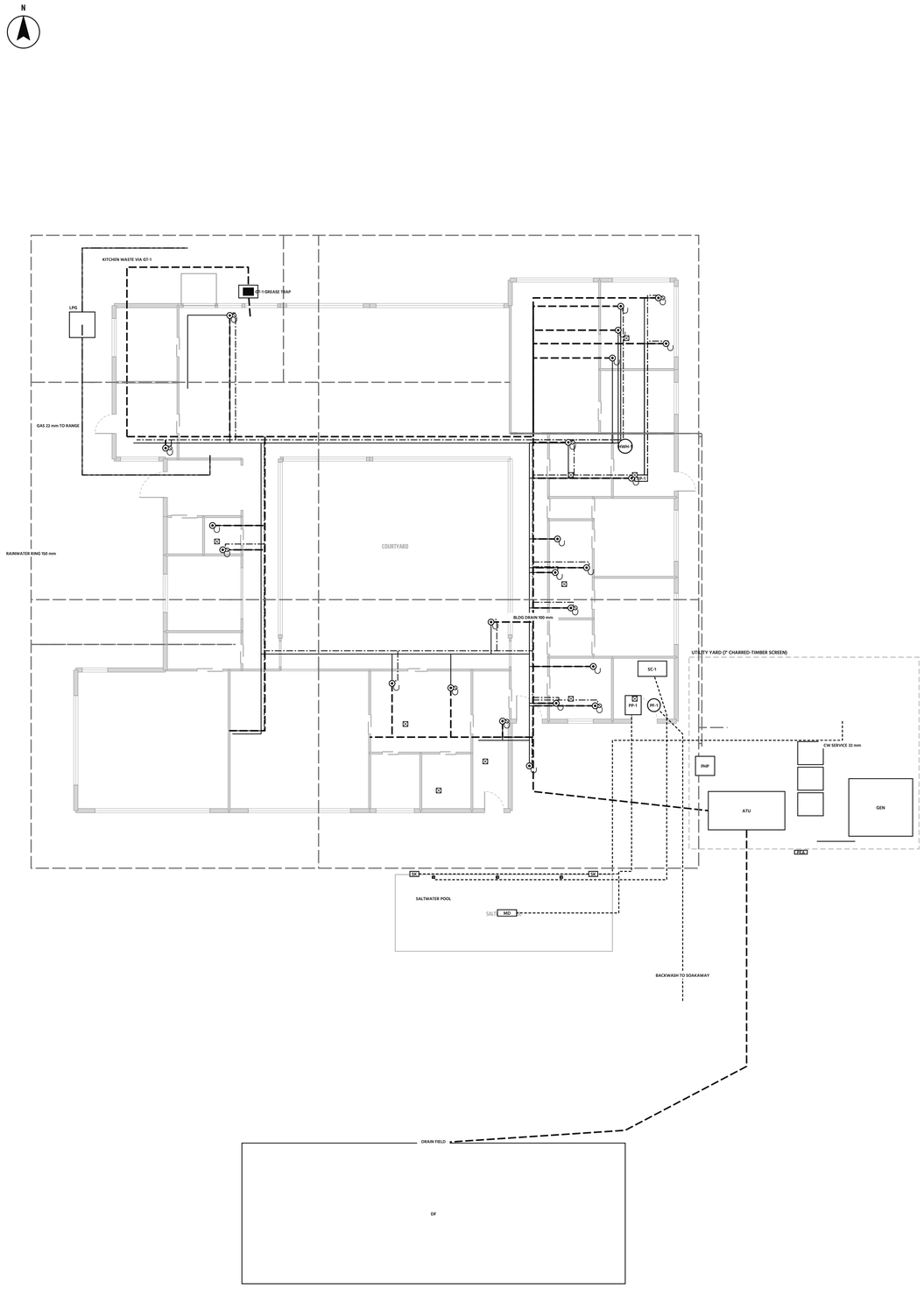
KITCHEN (ROUND 1)

Pro gas range 16 A (igniters); flush induction 20 A; steam + wall oven 32 A 2P; built-in stainless fridge 16 A; hood blower + MUA 16 A.

GFCI counter outlets at the appliance garage and island pop-ups; linear suspended LED on DALI. Detail on A-401.

SHEET TITLE
Power + Lighting Plan

SHEET
E-101



PLUMBING LEGEND

- Domestic cold water (PPR)
- Domestic hot water + recirculation
- Sanitary drainage (uPVC, under slab)
- Vent stack through roof
- Rainwater leader + collector
- Pool suction / return / fill
- Plumbing fixture

FIXTURE SCHEDULE

FIXTURE	CONNECTION	QTY
Cold Plunge (Chiller + Filter)	Auto-fill + 50 mm drain; P-trap at fixture, vented	1
Courtyard Shower	Cold + hot, to rain garden; P-trap at fixture, vented	1
Dog Wash Basin	50 mm drain with hair trap; P-trap at fixture, vented	1
Hinoki Soaking Tub	50 mm drain, 20 mm fill; P-trap at fixture, vented	1
Integrated Quartzite Sink + Dishwasher + Kitchen RO Tap	50 mm drain to grease trap; RO-1 under sink; P-trap at fixture, vented	1
Lavatory	40 mm drain, 15 mm supplies; P-trap at fixture, vented	5
Outdoor Shower	Cold + hot, to rain garden; P-trap at fixture, vented	1
Shower (Linear Drain)	50 mm drain, thermostatic valve; P-trap at fixture, vented	4
Steam Generator Feed	15 mm cold + drain; P-trap at fixture, vented	1
Tub-Shower	50 mm drain; P-trap at fixture, vented	1
Utility Sink	40 mm drain; P-trap at fixture, vented	1
Washer Box	50 mm standpipe; P-trap at fixture, vented	1
Water Closet (Wall-Hung, Dual Flush)	3/6 L, 100 mm drain; P-trap at fixture, vented	4

EQUIPMENT SCHEDULE

TAG	EQUIPMENT	SPECIFICATION
HWH-1	Heat-Pump Water Heater HWH-1	300 L, 1.5 kW heat pump with 3 kW backup element
RP-1	Recirculation Pump RP-1 + Manifold	Timer + thermostat control, 15 mm
GT-1	Grease Interceptor GT-1	100 L, precast, lid at grade
PP-1	Pool Pump (Variable Speed)	1.5 kW, VS pump, 50 mm suction / return
PF-1	Pool Media Filter	600 mm glass-media filter with 6-way valve
SC-1	Salt Chlorine Generator Cell + Controller	Sized for 60 m³, 1,000–4,000 ppm
PHP	Pool Heat Pump (Optional)	13 kW, in utility yard
RI	Pool Return Inlet	40 mm
RI	Pool Return Inlet	40 mm
RI	Pool Return Inlet	40 mm
SK	Skimmer	Wide-mouth
SK	Skimmer	Wide-mouth
MD	Main Drains (Dual, Anti-Entrapment)	2 x VGB-compliant covers
ATU	Septic Tank + Aerobic Treatment Unit	3,000 L septic + 2,000 L ATU (TIS 2550 type), lids at grade
DF	Drain Field (Leach Trenches)	60' x 22', 4 trenches, perc test required
GEN	Generator + Battery Pad	20 kVA diesel standby (auto-start) + 2 x 13.5 kWh LFP batteries, hybrid inverter
PEA	PEA Meter Pedestal + Main Isolator	3-phase 400 V, 30/100 A meter, surge protection

TRAPS, VENTS + GREASE

- Trapped floor drain FD (deep seal + trap primer)
- P-trap at every fixture
- Vent stack to open air above roof
- Grease interceptor GT-1 (kitchen, before septic)

1 **PLUMBING + SITE UTILITIES PLAN**

SCALE: 3/32" = 1'-0"

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
3/32" = 1'-0"

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

PLUMBING NOTES

Supply: well + rainwater, treated (sediment, carbon, UV, softener), booster pump 3 bar.

Sanitary to septic + ATU (TIS 2550 effluent) and drain field; perc test required.

Saltwater backwash and pool drain go to a lined soakaway, never to septic or garden.

Kitchen through grease interceptor GT-1 (Thai Building Control Act).

KITCHEN GAS + WATER

LPG 2 x 48 kg cage, auto-changeover regulator, 22 mm line in sleeve to the 48" range (wok burner) with leak-interlocked solenoid. Optional 15 mm branch to the outdoor wok station.

Integrated quartzite sink: 50 mm waste to grease trap GT-1; RO-1 drinking tap + fridge ice line. See A-401.

STANDARD

PPR fusion-welded supply, pressure test 1.5 x WP (min 10 bar, 2 h).

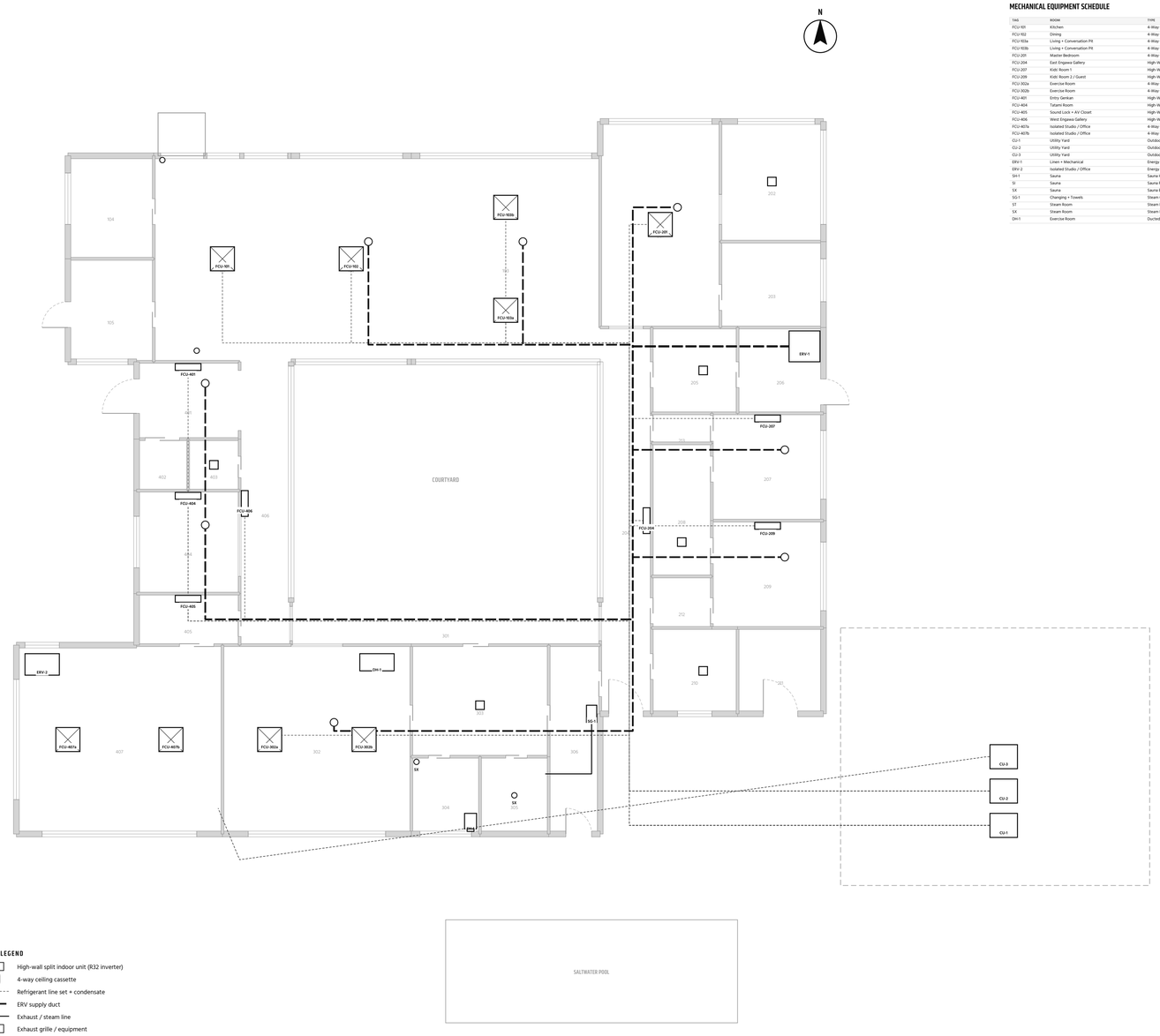
Waterproofing membrane in all wet areas; central heat-pump hot water.

SHEET TITLE

Plumbing + Site Utilities Plan

SHEET

P-101



MECHANICAL EQUIPMENT SCHEDULE

TAG	ROOM	TYPE	CAPACITY / SPEC	SYSTEM
FCU-101	Kitchen	4-Way Ceiling Cassette (Inverter, R32)	36,000 BTU/h (10.6 kW)	Multi-Split / VRF CU-1
FCU-102	Dining	4-Way Ceiling Cassette (Inverter, R32)	30,000 BTU/h (8.8 kW)	Multi-Split / VRF CU-1
FCU-102a	Living + Conversation Pit	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-1
FCU-102b	Living + Conversation Pit	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-1
FCU-201	Master Bedroom	4-Way Ceiling Cassette (Inverter, R32)	30,000 BTU/h (8.8 kW)	Multi-Split / VRF CU-2
FCU-204	East Engawa Gallery	High-Wall Split Indoor Unit (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-2
FCU-207	Kids' Room 1	High-Wall Split Indoor Unit (Inverter, R32)	18,000 BTU/h (5.3 kW)	Multi-Split / VRF CU-2
FCU-209	Kids' Room 2 / Guest	High-Wall Split Indoor Unit (Inverter, R32)	18,000 BTU/h (5.3 kW)	Multi-Split / VRF CU-2
FCU-302a	Exercise Room	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-2
FCU-302b	Exercise Room	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-2
FCU-401	Entry Genkan	High-Wall Split Indoor Unit (Inverter, R32)	9,000 BTU/h (2.6 kW)	Multi-Split / VRF CU-1
FCU-404	Tatami Room	High-Wall Split Indoor Unit (Inverter, R32)	12,000 BTU/h (3.5 kW)	Multi-Split / VRF CU-1
FCU-405	Sound Lock + AV Closet	High-Wall Split Indoor Unit (Inverter, R32)	9,000 BTU/h (2.6 kW)	Multi-Split / VRF CU-1
FCU-406	West Engawa Gallery	High-Wall Split Indoor Unit (Inverter, R32)	18,000 BTU/h (5.3 kW)	Multi-Split / VRF CU-1
FCU-407a	Isolated Studio / Office	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-3
FCU-407b	Isolated Studio / Office	4-Way Ceiling Cassette (Inverter, R32)	24,000 BTU/h (7.0 kW)	Multi-Split / VRF CU-3
CU-1	Utility Yard	Outdoor Heat-Pump Unit CU-1	162,000 BTU/h connected (47.5 kW), inverter VRF/multi-split	Multi-Split / VRF CU-1
CU-2	Utility Yard	Outdoor Heat-Pump Unit CU-2	150,000 BTU/h connected (43.9 kW), inverter VRF/multi-split	Multi-Split / VRF CU-2
CU-3	Utility Yard	Outdoor Heat-Pump Unit CU-3	48,000 BTU/h connected (14.1 kW), inverter VRF/multi-split	Multi-Split / VRF CU-3
ERV-1	Linen + Mechanical	Energy Recovery Ventilator ERV-1	350 m³/h, enthalpy core, MERV 13 + PM2.5 filter	Ventilation (ERV)
ERV-2	Isolated Studio / Office	Energy Recovery Ventilator ERV-2 (Studio)	120 m³/h, acoustic-lined ducts	Ventilation (ERV)
SH-1	Sauna	Sauna Heater SH-1	9 kW electric rock heater, 400 V 3-phase, guard rail	Sauna
SI	Sauna	Sauna Fresh-Air Inlet (Low, Behind Heater)	100 mm, from crawl / exterior	Sauna
SX	Sauna	Sauna Exhaust (High, Opposite Wall) + Fan	100 mm, 2 air changes/h, to outside	Sauna
SG-1	Changing + Towels	Steam Generator SG-1	9 kW, 400 V 3-phase, auto-flush, in Changing Room cabinet	Steam
ST	Steam Room	Steam Line to Steam Head	15 mm copper, sloped to head, insulated	Steam
SX	Steam Room	Steam Room Exhaust Fan (Post-Use Purge)	40 L/s, timer, to outside	Steam
DH-1	Exercise Room	Ducted Dehumidifier DH-1	90 L/day, for wet-room + gym humidity	Dehumidification

HVAC LEGEND

- High-wall split indoor unit (R32 inverter)
- 4-way ceiling cassette
- Refrigerant line set + condensate
- ERV supply duct
- Exhaust / steam line
- Exhaust grille / equipment

1 HVAC PLAN

SCALE: 1/8" = 1'-0"

WEBERMAN RESIDENCE

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Owner: Max Weberman

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MODEL
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UNITS
Feet + inches (mm for members)

REVISION
DD-01

HVAC NOTES

Cooling sized at about 850 BTU/h per m² (Thai rule of thumb for glazed rooms); verify with a heat-load calc.

ERV runs continuously for fresh air + humidity control; bath boost on humidity sensor.

Sauna: low inlet behind heater, high exhaust opposite. Steam: vapor-sealed, sloped ceiling, purge fan.

SHEET TITLE
HVAC Plan

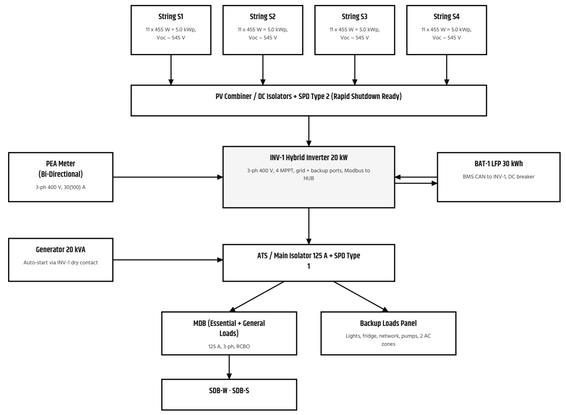
SHEET
M-101



1 **PV ROOF LAYOUT**

SCALE: 3/32" = 1'-0"

SINGLE-LINE DIAGRAM



Earthing: TN-C-S, PV frames bonded, separate DC SPD earth; all to < 5 ohm ring.

ANNUAL ENERGY ESTIMATE

LOAD	KWH/YR	BASIS
Split / VRF air-conditioning	16,400	Connected ~348k BTU/h (~25 kW elec); mountain climate, zoned scheduling
Pool pump (VS) + salt cell	2,600	0.6 kW avg x 10 h + cell
Sauna heater 9 kW	1,700	3 sessions/week incl. pre-heat
Steam generator 9 kW	750	2 sessions/week
Cold plunge chiller	1,100	~3 kWh/day
Heat-pump water heater	900	~2.5 kWh/day
Kitchen, laundry, lighting, plugs, ERV, studio, network, pumps	6,600	~18 kWh/day
Water: well, booster, treatment, irrigation pumps	500	
Total (Without EV)	30,550	~83 kWh/day avg, hot-season days ~110 kWh
Optional EV (15,000 km/yr)	2,400	7.4 kW charger, solar-timed

Peak demand (coincident) about 32 kW: AC 17 kW + sauna 9 kW + range 4 kW + misc; with EV ~40 kW.

OPTIONS (ROUGH THB, 2026)

OPTION	KWP	BATTERY	KWH/YR GEN	SELF-USED	COVERS	COST THB	SAVES THB/YR	PAYBACK
A: 12 kWp + 15 kWh	12	15 kWh	16,560	15,235	50%	570,000	71,000	8.0 yr
B: 20 kWp + 30 kWh (Recommended)	20	30 kWh	27,600	23,460	77%	1,020,000	115,000	8.9 yr
B0: 20 kWp, no battery	20	0 kWh	27,600	15,180	50%	600,000	77,000	7.8 yr
C: Full roof ~125 kWp + 30 kWh	125	30 kWh	172,500	25,875	85%	3,670,000	126,000	29.2 yr

Assumptions: yield 1380 kWh/kWp/yr on ~4 degree roofs; tariff 4.5 THB/kWh (PEA residential incl. Ft); export credit 2.2 THB/kWh only for the first 10 kWp under the PEA/ERC residential rooftop scheme (10-yr contract), surplus above that self-consumed or curtailed (zero-export). Installed cost ~30k THB/kWp incl. hybrid inverter; LFP ~14k THB/kWh.

Recommendation: Option B, 20 kWp on the south-facing planes (master roof + living north half) plus the east wing, with a 30 kWh battery: about 77% of annual use from solar, 9-yr payback, and 6-10 h of backup for essentials during mountain grid outages. The full roof (C) exports mostly unpaid power and never pays back. The north-facing south-wing and studio roofs stay clear (lower yield; sauna/steam vents).

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11 Oct 2026

SCALE
3/32" = 1'-0"

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

SOLAR NOTES

Clamp-on mounts to standing seam: no roof penetrations; keep 0.5 m from eaves for wind uplift.

Permits: PEA grid-connection approval, ERC exemption (< 1 MW), local building notice; engineer-signed roof check.

Tax deduction for residential rooftop solar (2025 measure, up to 200k THB) to be confirmed.

SHEET TITLE

Solar PV Plan + Single-Line Diagram

SHEET

E-102

CONTROLS + MONITORING BLOCK DIAGRAM

ENERGY

INV-1 Hybrid Inverter
Modbus RS485: PV, grid, load, surplus

BAT-1 LFP 30 kWh
SoC, reserve 30%

Per-Circuit CT Monitor
32 ch at MDB + SDB-W

Generator
Auto-start, run hours

WATER

Tank Levels
T-1, T-2, T-3, RT-1, TT-1

Well Pump Controller
Flow, dry-run, run hours

Treatment
Filter ΔP, backwash, UV lamp status

Booster BP-1
Pressure, faults

GARDEN

Irrigation Controller
6 drip zones, IP-1 pump

Soil Moisture x 6
One per zone

Weather Station
Rain, ET, wind, irradiance

Cistern Top-Up Valve
Well via air gap

Home Controller (Home Assistant) + Zigbee/Thread + UPS
Local-first in AV rack 40S; VPN for remote; nightly backup

Phone Dashboard + Genkan Wall Tablet
Energy · Water · Pool · Climate · Alerts



PHONE DASHBOARD CONCEPT

POOL + WELLNESS

Pool Pump VS + Salt Cell
Solar-surplus schedule

Sauna + Steam
Pre-heat from app, interlocks

Cold Plunge Chiller
Off-peak / surplus

CLIMATE

VRF Gateway CU-1/2/3
Setpoints, schedules, occupancy

ERV-1, ERV-2
Boost on humidity/CO2

Dehumidifier DH-1
Gym + wet room

SAFETY + LEAK

Leak Sensors x 14
Sinks, WCs, heater, washer, AHU

Auto Shut-Off Valve
Main at 20E + flow meter

Smoke / CO + Cameras
NVR in rack

KEY AUTOMATIONS

- Pool pump + salt cell run at 10:00-15:00 when PV surplus > 2 kW; minimum 6 h turnover guaranteed
- Battery reserve 30% for outages; generator auto-start below 15% or on long outage
- Irrigation: skip if rain > 3 mm forecast/24 h or soil moisture above threshold; water at 05:00 from cisterns, well top-up only if T-1,3 < 10%
- Leak detected: close SOV, stop BP-1, push alert; daily micro-leak check by flow meter at 03:00
- Filter ΔP high: schedule backwash on PV surplus; UV lamp fault: close SOV to house + alert
- AC: zone schedules by occupancy, pre-cool on solar surplus, setback 27 °C when away; sauna/steam pre-heat from the app
- Energy: per-circuit dashboard, monthly report, alert on abnormal baseload

CONTROLS BUDGET (ROUGH THB)

TAG	ITEM	THB
HUB	Home Controller (Home Assistant on Mini-PC) + Zigbee/Thread + UPS	30,000
CTM	Per-Circuit Energy Monitor (CT Clamps, 32 ch)	28,000
INVG	Inverter + Battery Gateway (Modbus RS485)	6,000
AVG	AC/VRF Gateway (BACnet/Modbus)	25,000
POOL	Pool Pump + Salt Cell Controller	20,000
IRR	Smart Irrigation Controller (6 Zones)	12,000
WX	Weather Station (Rain, Wind)	12,000
LVL1	Level Sensors (T-1, T-2, T-3, RT-1, TT-1)	25,000
SM	Soil Moisture Sensors (6)	12,000
WP	Well Pump Controller + Flow Meter	15,000
DP	Filter ΔP Sensors + UV Status Relay	14,000
SOV	Motorized Auto Shut-Off Valve + Flow Meter	18,000
LK	Leak Sensors (14)	17,000
DASH	Wall Dashboard Tablet (Genkan) + Phone App	15,000
PRG	Wiring, Integration + Programming	90,000
	Total	339,000

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE
Design Development

NOT FOR CONSTRUCTION

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CONTROLS NOTES

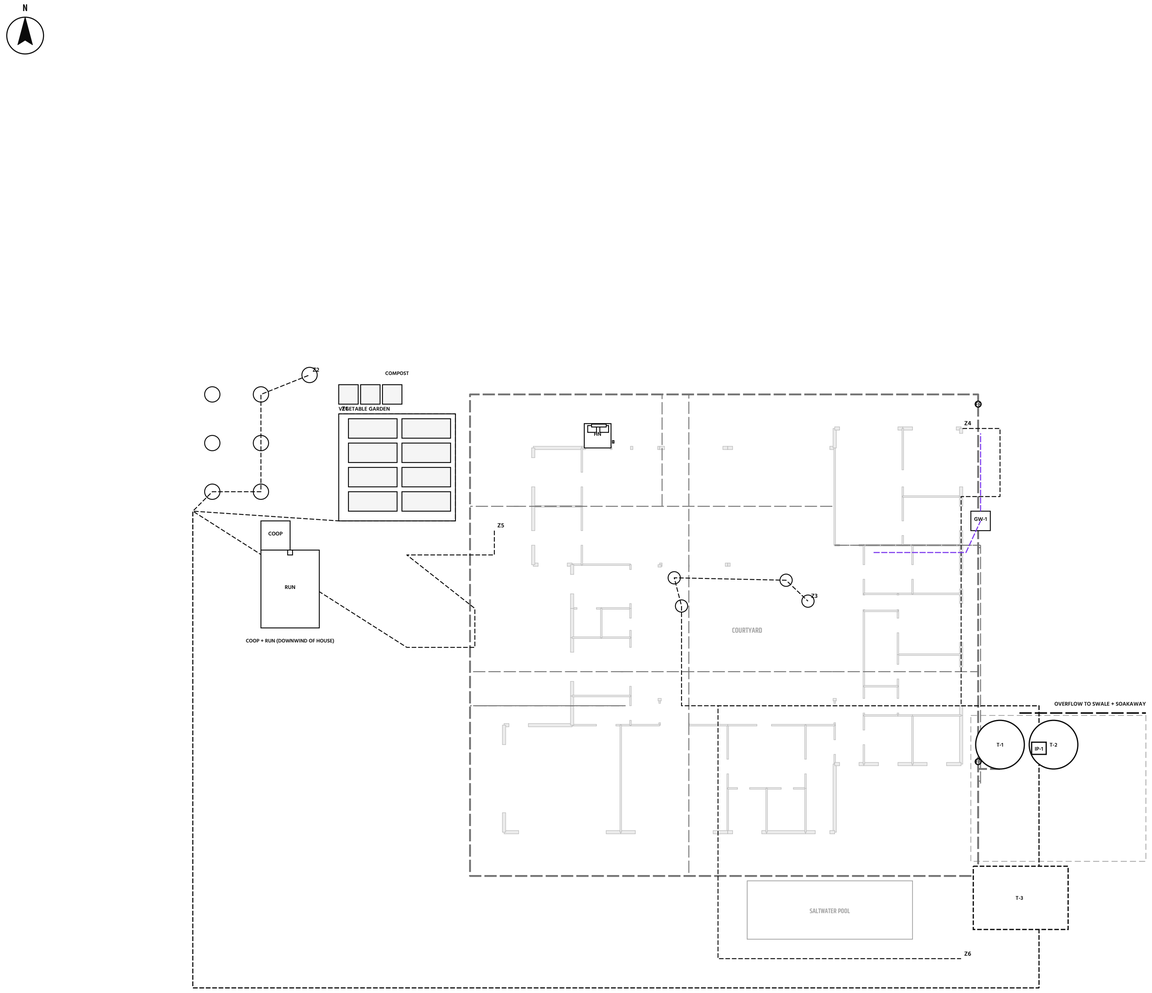
Local-first: automations keep running without internet.

Open protocols (Modbus, Zigbee, Matter) so devices can be swapped.

Device locations shown on the Controls layer in the 3D viewer.

SHEET TITLE
Controls + Monitoring

SHEET
E-103



1 **RAINWATER, IRRIGATION + KITCHEN GARDEN PLAN**

SCALE: 1/16" = 1'-0"

CISTERN SIZING

Roof catchment 815 m² (all planes incl. eaves); about 35% (living, master, east) piped to the garden cisterns, the rest to swales. Rainfall about 1,207 mm/yr, ~88% May-Oct (monsoon). Runoff factor 0.8 (metal roof, first flush). Dry-season (Nov-Apr) drip demand 142 m³ for the vegetable garden (49 m²), 4 maples, 7 orchard trees and ~120 m² planting.

MONTH	INFLOW M³	DEMAND M³	SUPPLIED M³	STORED M³
Oct	27.4	8	8	72.0
Nov	9.1	18	18	62.0
Dec	2.7	22	22	42.7
Jan	1.8	24	24	20.6
Feb	2.7	24	23.3	0.0
Mar	5.7	27	5.7	0.0
Apr	13.7	27	13.7	0.0
May	36.5	10	10	26.5
Jun	34.2	4	4	56.7
Jul	38.8	4	4	76.0
Aug	50.2	3	3	77.0
Sep	52.5	3	3	77.0

OPTIONS

TOTAL STORAGE	DRY-SEASON COVERAGE	ROUGH THB (INCL. GUTTERS, FF, PUMP, DRIP)
38 m³	46%	415,000
60 m³	61%	570,000
80 m³	75%	710,000
120 m³	100%	990,000

Selected: 80 m³ = T-1 + T-2 (2 x 19 m³, utility yard) + buried T-3 (42 m³ crate tank). Covers about 75% of dry-season irrigation; the balance (Jan-Apr) is topped up from the well through an air gap, controlled by tank level + soil moisture.

LEGEND

- Drip zone lateral (Z1-Z6)
- Rainwater collector + leaders
- Buried cistern
- Tree / maple (bubblers)

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RAINWATER NOTES

Gutters on every low eave with leaf screens; first-flush diverters (~1 mm/m²) at both ring inlets.

Garden water is non-potable: purple pipe, labels, no cross-connection with house supply.

Overflow to a planted swale + soakaway down-slope, away from the drain field.

SHEET TITLE

Rainwater, Irrigation + Kitchen Garden Plan

SHEET

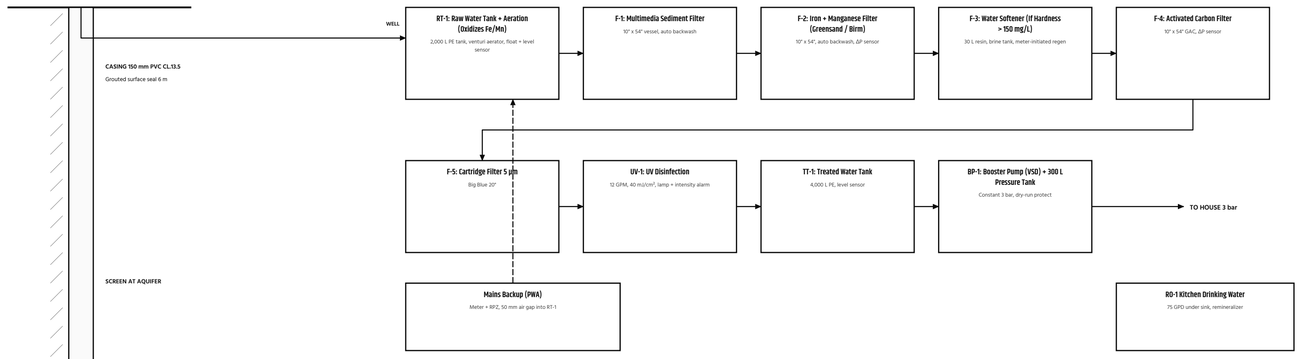
P-102

WATER SOURCE PRIORITY + TREATMENT SCHEMATIC

1. Rainwater
Garden cisterns T-1/T-2/T-3 for irrigation only

2. Drilled Well
Primary house supply, via treatment

3. Mains (PWA)
Backup into RT-1 via 50 mm air gap + RPZ



1 **TYPICAL DEEP WELL**
SCALE: NTS

SIZING + PRIORITY

Demand: 6 people x 200 L + pool/plunge make-up ≈ 1.5 m^3/day; peak 60 L/min
Well depth: 60-120 m (typical Thai deep well), confirm by DGR hydro-survey
Casing: 150 mm (6") PVC Class 13.5 + steel surface casing, grouted 6 m
Pump: Submersible 1.5 kW (2 HP), 3 m^3/h @ 60 m head, dry-run + low-level cut-out

Storage: 2,000 L raw (aeration, contact time) + 4,000 L treated (≈ 2.5 days) + 300 L pressure tank. Cisterns T-1,T-3 stay garden-only.

WATER TEST FIRST: Lab water test (Fe, Mn, hardness, TDS, pH, nitrate, arsenic, coliform) BEFORE final treatment selection; about 3,000-5,000 THB. Iron/manganese and softening stages are installed only if the test shows they are needed.

Controls: well pump on RT-1 level; ΔP alerts call a backwash on solar surplus; UV fault closes the house valve and alerts; tank levels on the dashboard (E-103).

ROUGH COST (THB)

ITEM	THB
Drilling 100 m incl. casing, gravel pack, grout, development	180,000
Submersible pump, controller, cable, wellhead	40,000
Treatment train (RT-1 to BP-1)	251,000
Kitchen RO	15,000
Mains backup, RPZ, air gaps, piping, install	70,000
Water test	5,000
Total	561,000

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WATER NOTES

Well permit from the DGR (Groundwater Act) for wells > 15 m deep; licensed driller.

Backflow: RPZ on mains, air gaps on every top-up, no cross-connection with garden water.

Locate the well > 30 m from the drain field and septic.

SHEET TITLE
Water Source + Treatment

SHEET
P-103